

Multiple Choice (10 marks)

1. Which of the following is least likely to reduce bias in a sample survey?
 - (a) Following up with those who did not respond to the survey the first time
 - (b) Asking questions in a neutral manner to avoid influencing the responses
 - (c) Using stratified random sampling rather than simple random sampling
 - (d) Selecting samples randomly
2. How many years are there in a decade?
 - (a) 5
 - (b) 10
 - (c) 15
 - (d) 20
3. The domain of the function $h(x) = \sqrt{25 - x^2} + \sqrt{-(x - 2)}$ is an interval of what width?
 - (a) 7
 - (b) 5
 - (c) 2
 - (d) 25
4. Members of the Hiking Club hiked for 2.4 hours covering 2.5 miles each hour. How far did the members hike
 - (a) 4.9 miles
 - (b) 5.8 miles
 - (c) 6.0 miles
 - (d) 6.3 miles
5. John has five children. What is the probability that at least half of them are girls? (We can assume a boy is equally likely to be born as is a girl, and vice-versa.)
 - (a) $\frac{1}{2}$
 - (b) $\frac{1}{4}$
 - (c) $\frac{1}{8}$
 - (d) $\frac{3}{5}$

True / False (10 marks)

Answer True or False

6. True or False: Dividing 942 by 3 gives a result of 214.
7. True or False: The sum of the center coordinates and radius of the circle C is -9.
8. True or False: The cost of a 6-lb bag of cat food at 0.47 per pound is 2.82.
9. True or False: The equation $36 - y = 13$ represents the same relationship as 36 less than a number y being equal to 13.
10. True or False: It takes the machine 30 minutes to fill 500 bottles of soda at the same rate.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Henry's Hamburger Heaven offers its hamburgers with the following condiments: ketchup, mustard, mayonnaise, tomato, lettuce, pickles, cheese, and onions. A customer can choose one, two, or three meat patties, and any collection of condiments. How many different kinds of hamburgers can be ordered?
12. Explain how you would express the phrase "3 fewer than a number, p" in mathematical terms. Discuss the reasoning behind your expression and its components.

End of Paper